The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN/SPRING SEMESTER 2023-2024

Mathematics for Computer Scientists

Time allowed: 2 Hours and 0 Minutes

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer all FOUR questions. All questions are worth 25 marks each, hence the total mark is 100.

No calculators are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

ADDITIONAL MATERIAL: None.
INFORMATION FOR INVIGILATORS: None.

Question 1: This question is about *Logic, Proof, and Functions*.

[**overall 25 marks**]

a. Use a truth table to prove or disprove the following statements:

(i) $\neg(p \vee (q \wedge r)) \equiv (\neg p) \wedge (\neg q \vee \neg r)$

(ii) $\neg(p \wedge (q \vee r)) \equiv (\neg p) \vee (\neg q \vee \neg r)$

(4 marks)

b. Prove $\forall x(W(x) \rightarrow \neg Q(x)$ is true, given the following two premises are true:

$\neg\exists x(Q(x) \wedge \neg H(x))$
$\forall x(W(x) \rightarrow \neg H(x)$

Please use the format of "Step-Reason" for your proof. (4 marks)

c. Translate the following two sentences from English to predicate logic. Suppose the domain that you are working over is X, the set of people. Let $F(x, y)$ denote 'x and y are friends'. Then use De Morgan's laws to express their negation such that no negation operator precedes a quantifier.

(i) Everyone has a friend. (2 marks)

(ii) Someone has exactly one friend. (6 marks)

d. Use induction to prove that for all integers $n \geq 0$,

$$F_0 + F_1 + F_2 + \cdots + F_n = F_{n+2} - 1$$

where F_n is the n_{th} Fibonacci number $(F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2}$ for $n > 1)$.

(9 marks)

Question 2: This question is about *Set, Functions, Relations, and Probability*.

[**overall 25 marks**]

a. Let $f(x)$ be a function from $\mathbb{R}$ to $\mathbb{R}$. Is the function $f(x) = \frac{x}{x^2+1}$ invertible? If yes, what is its inverse? If not, justify your answer.

[5 Marks]

b. For all sets A and B, prove or disprove $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$, where $\mathcal{P}(X)$ is the power set of X.

[5 Marks]

c. Let A be the set of all propositional statements. A relation $\mathbb{R}$ is defined on A as follows: For all $p, q \in A$,

$$\{(p, q) \in \mathbf{R} \mid p \rightarrow q \text{ is true.}\}$$

Determine whether the relation is reflexive, symmetric, anti-symmetric, transitive or none of these. Justify your answer.

(8 Marks)

d. A bag contains 3 red and 7 black balls. Two balls are drawn at randon without replacement. If the second ball is red, what is the probability that the first ball is also red? (7 marks)

Question 3: This question is about *Optimization and Matrix Rank*.

[**overall 25 marks**]

a. A cylindrical container is built to hold a fixed volume v cubic meters. It needs to be constructed to minimize the cylinder's total inner surface area, which includes its lid and base.

- Denote the height of the cylinder as h meters.
- Denote the radius of the circular base and lid of the cylinder as r meters.

To complete this question part, you will need the following formulae:

- Area of a circle of radius r is πr^2.
- Circumference (perimeter) of a circle of radius r is $2\pi r$.
- The volume of the cylinder is $v = \pi r^2 h$.

(i) Show that the total inner surface area of the cylinder is

$$2\pi r^2 + 2vr^{-1}.$$

(3 marks)

(ii) For which value of r is the surface area of the cylinder minimized for a fixed volume v? (5 marks)
(Write your answer in terms of π and v).

(iii) Show that the ratio of height to radius, h/r, is always 2, when the surface area of the cylinder is minimized. (2 marks)

b. Answer these questions related to ranks of matrices.

(i) Which of these four statements are true and which are false? (4 marks)

(1) A matrix is singular if and only if its determinant is zero.
(2) The rows of a matrix are linearly dependent if the linear combination of those rows is the identity matrix.
(3) The rank of a matrix is always greater than its order.
(4) If the rank of a square matrix is less than its order, it means its columns are linearly dependent.

(ii) Express the last column of this matrix as a linear combination of the first three columns:

$$\begin{pmatrix} 1 & 0 & -1 & -1 \\ -1 & -2 & 0 & 0 \\ 2 & 3 & 1 & 4 \end{pmatrix}$$

(2 marks)

(iii) Compute the rank of each of these three matrices, and show your working.
(6 marks)

$$A = \begin{pmatrix} 1 & 2 & 4 & 1 \\ 0 & 0 & 2 & 0 \\ 2 & 4 & 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 0 & 4 & -2 & -1 \\ 3 & 0 & 6 & -3 & 1 \end{pmatrix}$$

(iv) Which of the three matrices above have linearly dependent rows? (3 marks)

Question 4: This question is about *linear mappings*.

[**overall 25 marks**]

a. Which of these mappings are linear, which are affine, and which are neither?

- $f_1 : \mathbb{R} \to \mathbb{R}, \quad f_1(x) = 3x + 1$
- $f_2 : \mathbb{R}^2 \to \mathbb{R}, \quad f_2(x, y) = xy + x + y$
- $f_3 : \mathbb{R}^3 \to \mathbb{R}^2, \quad f_3(x, y, z) = (x, y)$
- $f_4 : \mathbb{R} \to \mathbb{R}^3, \quad f_4(x) = (2x, 3x, 0)$

(4 marks)

b. Derive the vector space for each of these bases in the form $\{(x, y, z) \in \mathbb{R}^3 \mid \cdots = 0\}$ (i.e. complete the dots $\cdots$). Show your working.

(i) $(1, 0, 0),\ (1, 2, 0)$
(ii) $(\frac{1}{3}, 0, 1),\ (\frac{2}{3}, 1, 0)$
(iii) $(1, 1, -1),\ (0, 0, 2)$

(6 marks)

c. Consider the mapping, $f : \mathbb{R}^2 \to \mathbb{R}^2, f(x, y) = (x + 2y, x)$, and the sets $U = \{(x, y) \in \mathbb{R}^2 \mid x = y\}$ and $W = \{(x, y) \in \mathbb{R}^2 \mid y = 0\}$.

(i) Compute the image of U under f, $f(U)$. (2 marks)
(ii) Compute the inverse image of W through f, $f^{-1}(W)$. (2 marks)

d. Let $f : E \to F$ be a linear mapping. Prove the following theorem:

If $(W, +, \cdot)$ is a vector subspace of $(F, +, \cdot)$, then $\left(f^{-1}(W), +, \cdot\right)$ is a vector subspace of $(E, +, \cdot)$.

(5 marks)

e. Let $f : E \to F$ be a linear mapping. Give the definition of the kernel of f, ker(f). (2 marks)

f. Let $f : \mathbb{R}^3 \to \mathbb{R}^3, f(x, y, z) = (x + 2y, 2y + z, x - z)$.

(i) Compute ker(f). (3 marks)
(ii) Compute the dimension, dim(ker(f)). (1 mark)